

3.2 Worksheet

1. Determine whether each of the following series converge or diverge. If they converge, find what value they converge to.

a) $\sum_{n=1}^{\infty} \frac{7}{2^{n+2}}$

We can see that $\sum_{n=1}^{\infty} \frac{7}{2^{n+2}} = \sum_{n=1}^{\infty} \frac{7}{2^2} \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} \frac{7}{4} \cdot \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{7}{4} \cdot \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} = \sum_{n=1}^{\infty} \frac{7}{8} \left(\frac{1}{2}\right)^{n-1}$. This is a geometric series with first term $\frac{7}{8}$ and common ratio $r = \frac{1}{2}$. Since $|r| < 1$ this geometric series converges,

specifically $\sum_{n=1}^{\infty} \frac{7}{2^{n+2}} = \sum_{n=1}^{\infty} \frac{7}{8} \left(\frac{1}{2}\right)^{n-1} = \frac{7/8}{1-1/2} = \frac{7/8}{1/2} = \frac{7}{4}$

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b) $\sum_{k=0}^{\infty} 2^{2k} \cdot 3^{1-k}$

We can see that $\sum_{k=0}^{\infty} 2^{2k} \cdot 3^{1-k} = \sum_{k=0}^{\infty} \frac{2^{2k}}{3^{k-1}} = \sum_{k=0}^{\infty} \frac{4^k}{3^k \cdot 3^{-1}} = \sum_{k=0}^{\infty} 3 \cdot \frac{4^k}{3^k} = \sum_{k=0}^{\infty} 3 \cdot \left(\frac{4}{3}\right)^k$. This is a geometric series with first term 3 and common ratio $r = \frac{4}{3}$. Since $|r| \geq 1$ this geometric series diverges.

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c) $\sum_{k=0}^{\infty} \frac{3^k}{e^{k+1}}$

First we rewrite the series as $\sum_{k=0}^{\infty} \frac{3^k}{e^{k+1}} = \sum_{k=0}^{\infty} \frac{3^k}{e^k \cdot e} = \sum_{k=0}^{\infty} \frac{1}{e} \cdot \left(\frac{3}{e}\right)^k$.

We note that this is a geometric series with $a = \frac{1}{e}$ & $r = \frac{3}{e}$. Since $|r| \geq 1$ this geometric series diverges.

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2. Each of following is a geometric series. Determine if this series converges or diverges. If it converges, then find its sum.

a) $3 - 4 + \frac{16}{3} - \frac{64}{9} + \dots$

We see that each is being multiplied by $-\frac{4}{3}$ in order to create the next term in the series. Therefore

$r = -\frac{4}{3}$ for this geometric series and clearly $a = 3$. Therefore, we have

$$3 - 4 + \frac{16}{3} - \frac{64}{9} + \dots = \sum_{n=1}^{\infty} 3 \left(-\frac{4}{3} \right)^{k-1}. \text{ Since } |r| \geq 1 \text{ this series diverges.}$$

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b) $4 + 3 + \frac{9}{4} + \frac{27}{16} + \dots$

We see that each is being multiplied by $\frac{3}{4}$ in order to create the next term in the series. Therefore

$$r = \frac{3}{4} \text{ for this geometric series and clearly } a = 4. \text{ Therefore, we have } 4 + 3 + \frac{9}{4} + \frac{27}{16} + \dots = \sum_{n=1}^{\infty} 4 \left(\frac{3}{4} \right)^{k-1}.$$

$$\text{Since } |r| < 1 \text{ this series converges and so } 4 + 3 + \frac{9}{4} + \frac{27}{16} + \dots = \sum_{n=1}^{\infty} 4 \left(\frac{3}{4} \right)^{k-1} = \frac{4}{1 - \frac{3}{4}} = \frac{4}{\left(\frac{1}{4} \right)} = 16.$$

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c) $10 - 2 + 0.4 - 0.08 + \dots$

We see that each is being multiplied by $-\frac{1}{5}$ in order to create the next term in the series. Therefore

$r = -\frac{1}{5}$ for this geometric series and clearly $a = 10$. Therefore, we have

$$10 - 2 + 0.4 - 0.08 + \dots = \sum_{n=1}^{\infty} 10 \left(-\frac{1}{5} \right)^{k-1}. \text{ Since } |r| < 1 \text{ this series converges and so}$$

$$10 - 2 + 0.4 - 0.08 + \dots = \sum_{n=1}^{\infty} 10 \left(-\frac{1}{5} \right)^{k-1} = \frac{10}{1 - \left(-\frac{1}{5} \right)} = \frac{10}{\left(\frac{4}{5} \right)} = \frac{50}{4} = \frac{25}{2}.$$

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d) $2 + 0.5 + 0.125 + 0.03125 + \dots$

We see that each is being multiplied by $\frac{1}{4}$ in order to create the next term in the series. Therefore $r = \frac{1}{4}$ for this geometric series and clearly $a = 2$. Therefore, we have

$$2 + 0.5 + 0.125 + 0.03125 + \dots = \sum_{n=1}^{\infty} 2 \left(\frac{1}{4} \right)^{k-1}. \text{ Since } |r| < 1 \text{ this series converges and so}$$

$$2 + 0.5 + 0.125 + 0.03125 + \dots = \sum_{n=1}^{\infty} 2 \left(\frac{1}{4} \right)^{k-1} = \frac{2}{1 - \left(\frac{1}{4} \right)} = \frac{2}{\left(\frac{3}{4} \right)} = \frac{8}{3}.$$

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3. $\sum_{k=0}^{\infty} (\sqrt{k} - \sqrt{k+1})$

We look at the sequence of partial sums for this series:

$$s_n = (\sqrt{0} - \sqrt{1}) + (\sqrt{1} - \sqrt{2}) + (\sqrt{2} - \sqrt{3}) + \dots + (\sqrt{n} - \sqrt{n+1}) = \sqrt{0} - \sqrt{n+1} = -\sqrt{n+1}$$

We can see that $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} -\sqrt{n+1} = -\infty$.

Therefore, since the sequence of partial sums does not converge the series does not converge by definition. ■

4. $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$

We look at the sequence of partial sums for this series:

$$s_k = \left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{4} - \frac{1}{6} \right) + \dots + \left(\frac{1}{k-2} - \frac{1}{k} \right) + \left(\frac{1}{k-1} - \frac{1}{k+1} \right) = 1 + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1}$$

We can see that $\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \left(\frac{3}{2} - \frac{1}{k} - \frac{1}{k+1} \right) = \frac{3}{2}$.

Therefore since the sequence of partial sums converges to $\frac{3}{2}$ the series $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$ converges to

$\frac{3}{2}$ by definition, that is $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n+1} \right) = \frac{3}{2}$ ■

5. $\sum_{n=1}^{\infty} \left(\cos\left(\frac{1}{n^2}\right) - \cos\left(\frac{1}{(n+1)^2}\right) \right)$

We look at the sequence of partial sums for this series:

$$\begin{aligned} s_k &= \left(\cos(1) - \cos\left(\frac{1}{4}\right) \right) + \left(\cos\left(\frac{1}{4}\right) - \cos\left(\frac{1}{9}\right) \right) + \left(\cos\left(\frac{1}{9}\right) - \cos\left(\frac{1}{16}\right) \right) + \dots + \left(\cos\left(\frac{1}{k^2}\right) - \cos\left(\frac{1}{(k+1)^2}\right) \right) \\ &= \cos(1) - \cos\left(\frac{1}{(k+1)^2}\right) \end{aligned}$$

We can see that $\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \left(\cos(1) - \cos\left(\frac{1}{(k+1)^2}\right) \right) = \cos(1) - \cos(0) = \cos(1) - 1$.

Therefore since the sequence of partial sums converges to $\cos(1) - 1$ the series

$\sum_{n=1}^{\infty} \left(\cos\left(\frac{1}{n^2}\right) - \cos\left(\frac{1}{(n+1)^2}\right) \right)$ converges to $\cos(1) - 1$ by definition, that is

$$\sum_{n=1}^{\infty} \left(\cos\left(\frac{1}{n^2}\right) - \cos\left(\frac{1}{(n+1)^2}\right) \right) = \cos(1) - 1$$
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6. $\sum_{n=1}^{\infty} (e^{1/n} - e^{1/(n+1)})$

We look at the sequence of partial sums for this series:

$$s_k = (e^1 - e^{1/2}) + (e^{1/2} - e^{1/3}) + (e^{1/3} - e^{1/4}) + \dots + (e^{1/k} - e^{1/(k+1)}) = e - e^{1/(k+1)}$$

We can see that $\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} (e - e^{1/(k+1)}) = e - e^0 = e - 1$.

Therefore since the sequence of partial sums converges to $e - 1$ the series $\sum_{n=1}^{\infty} (e^{1/n} - e^{1/(n+1)})$ converges to

$$e - 1 \text{ by definition, that is } \sum_{n=1}^{\infty} (e^{1/n} - e^{1/(n+1)}) = e - 1$$

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7. $\frac{2}{1 \cdot 3} + \frac{2}{2 \cdot 4} + \frac{2}{3 \cdot 5} + \frac{2}{4 \cdot 6} + \frac{2}{5 \cdot 7} + \frac{2}{6 \cdot 8} + \dots$ (Hint: Try writing this series in summation notation)

We can express this series as $\sum_{n=1}^{\infty} \frac{2}{n(n+2)}$. Using partial fractions we see that

$$\frac{2}{n(n+2)} = \frac{A}{n} + \frac{B}{n+2} \text{ which implies } 2 = A(n+2) + Bn.$$

If $n = 0$ then $2 = 2A \Leftrightarrow A = 1$.

If $n = -2$ then $2 = -2B \Leftrightarrow B = -1$.

$$\text{Therefore } \sum_{n=1}^{\infty} \frac{2}{n(n+2)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right).$$

We now write a sequence of partial sums for this series, which will be telescoping.

$$\begin{aligned} s_n &= \left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{4} - \frac{1}{6} \right) + \dots + \left(\frac{1}{n-2} - \frac{1}{n} \right) + \left(\frac{1}{n-1} - \frac{1}{n+1} \right) + \left(\frac{1}{n} - \frac{1}{n+2} \right) \\ &= 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \end{aligned}$$

$$\text{We can see that } \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right) = 1 + \frac{1}{2} - 0 - 0 = \frac{3}{2}.$$

Therefore since the sequence of partial sums converges to $\frac{3}{2}$ the series $\sum_{n=1}^{\infty} \frac{2}{n(n+2)}$ converges to $\frac{3}{2}$ by

$$\text{definition, that is } \sum_{n=1}^{\infty} \frac{2}{n(n+2)} = \frac{3}{2}$$

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$$8. \sum_{n=1}^{\infty} \frac{n+1}{n^2+n}$$

We note that $\sum_{n=1}^{\infty} \frac{n+1}{n^2+n} = \sum_{n=1}^{\infty} \frac{n+1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n}$.

This is the harmonic series and so this series diverges. ■

$$9. \sum_{n=2}^{\infty} \frac{1}{n-1}$$

We note that this series is $\sum_{n=2}^{\infty} \frac{1}{n-1} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ and so this is the Harmonic series and it diverges. ■